Exercise Sheet 2

Probability, Data and a Parameter

Probabilistic Machine Learning

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2.A Conditioning and Bayes

Exercise 2.A.1: Consequences of the Axioms

Using only the three Kolmogorov axioms (and set algebra), show:

1. $P(A \setminus B) = P(A) - P(A \cap B)$
2. $P(A \cup B) \leq P(A) + P(B)$
3. If $A \subseteq B$, then $P(A) \leq P(B)$.

Exercise 2.A.2: One Flag, Two Flags

Fraud occurs in **0.2%** of transactions. A rule flags **95%** of frauds and **2%** of legitimate transactions.

1. What fraction of flagged transactions is fraud?
2. A second, different rule has the same rates. Assume the two rules err **independently given the class**. A transaction is flagged by both. Now?
3. When would the assumption in 2 be badly wrong — and in which direction would the answer then be off?

Exercise 2.A.3: Independent Is Not Disjoint

1. Show that two disjoint events A, B with $P(A), P(B) > 0$ are never independent.
2. Flip two fair coins. Let A = "first is heads", B = "second is heads", C = "the coins agree". Show that A, B, C are pairwise independent but not mutually independent.

2.B Random Variables and the Joint

Exercise 2.B.1: App Use and Churn

The joint distribution of X (uses the app: 1/0) and Y (churns: 1/0):

	$Y = 0$	$Y = 1$
$X = 0$	0.30	0.20
$X = 1$	0.40	0.10

1. Compute $P(X = 1)$, $P(Y = 1)$, $P(Y = 1 \mid X = 1)$ and $P(Y = 1 \mid X = 0)$.
2. Are X and Y independent? Compute $\mathrm{cov}(X, Y)$.
3. Does the table show that the app *reduces* churn? What else could produce it?

Exercise 2.B.2: Dependent, Yet Uncorrelated

Let X take the values $-1, 0, 1$ with probability $\frac{1}{3}$ each, and $Y = X^2$.

1. Compute $E[X]$, $E[Y]$ and $\mathrm{cov}(X, Y)$.
2. Show that X and Y are not independent.
3. What does this say about using correlation to screen features?

2.C Data, Statistics and a Parameter

Exercise 2.C.1: Scale and Model

For each attribute of a customer table, give the scale and a first probabilistic model, and name one statistic that would be **invalid**: payment method (card / invoice / PayPal) · delivery rating (1–5 stars) · number of orders last year · order value in € · has a newsletter subscription.

Exercise 2.C.2: How Precise Is a Conversion Rate?

A campaign reached **400** users; **60** converted.

1. Estimate the conversion rate and its standard error.
2. Give an approximate 95% confidence interval and state correctly what the 95% refers to.
3. How many users would give a half-width of ± 0.01?

Exercise 2.C.3: Two Analysts, One Campaign

Same data: 60 of 400 converted. Analyst A uses a flat prior $\mathrm{Beta}(1, 1)$; analyst B, from earlier campaigns, uses $\mathrm{Beta}(20, 80)$.

1. Write down both posteriors and their means.
2. How many "pseudo-observations" is B's prior worth? Why do the two analysts nearly agree?
3. The data arrived in two batches: 25 of the first 200, 35 of the second 200. Show that updating twice gives the same posterior as updating once.
4. What is A's probability that the next user converts?

Exercise 2.C.4: A Sampling Distribution by Simulation (Python)

```python
import numpy as np
rng = np.random.default_rng(0)
samples = rng.exponential(scale=2.0, size=(1000, 25))   # 1000 samples, n = 25
```

1. Compute the mean of each sample and the standard deviation of those means. Compare with $\sigma/\sqrt{n}$ (for this distribution $\sigma = 2$).
2. The exponential is strongly skewed. Plot a histogram of the means — what shape do you see, and which theorem predicts it?
3. Repeat with $n = 100$. By what factor does the spread shrink?